

PHASE 1 — PROJECT DOCUMENTATION

University of Ghana Dental School · College of Health Sciences

UGDS Patient Feedback Survey Management System

Korle Bu Campus, Accra, Ghana

DOCUMENT VERSION

Version 1.0

DATE

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PREPARED BY

Customer Care & Digital Systems Unit

CLASSIFICATION

Internal Use Only



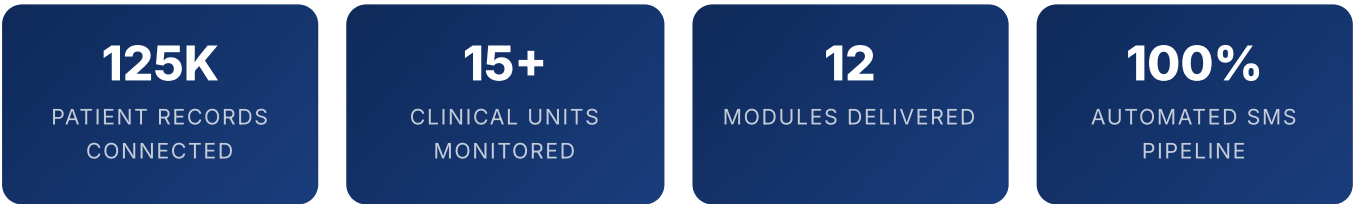
Deployed & Operationally Verified

Executive Summary

The **UGDS Patient Feedback Survey Management System** is a purpose-built, real-time clinical governance and patient engagement platform developed specifically for the University of Ghana Dental School. It bridges UGDS's physical archive of **125,042 patient records** with automated digital touchpoints — transforming how the institution captures, monitors, and acts on patient experience data.

This document provides a complete overview of all features delivered in **Phase 1**, with visual evidence from the live system, along with the roadmap for future development phases.

System at a Glance

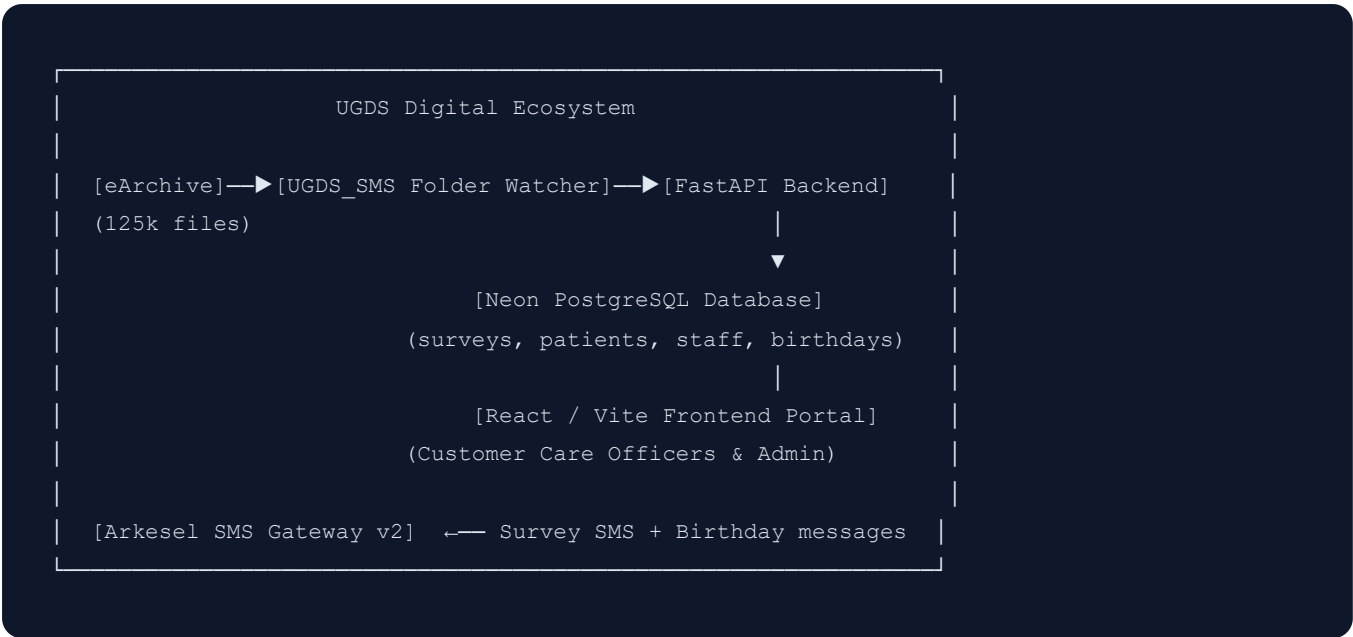


1. The Problem We Solved

Before this system, UGDS faced critical operational gaps that this platform directly addresses:

Challenge	Impact on UGDS
No structured way to collect patient feedback	Clinical quality blind spots
Patient records locked in physical folders	Patients could not be reached for follow-up
No visibility into which clinics were underperforming	Leadership had no data-driven governance tool
Birthday outreach was manual or non-existent	Missed patient relationship-building opportunities
No monthly reports formatted for the Dean's office	Decisions were based on opinion, not evidence

2. System Architecture



Frontend
React 18 + Vite, Vanilla CSS, Recharts for data visualization

Backend
FastAPI (Python) — RESTful API with secure endpoints

Database
Neon PostgreSQL — cloud-hosted, real-time data persistence

SMS Gateway
Arkesel API v2 — Ghana-based gateway for survey & birthday SMS

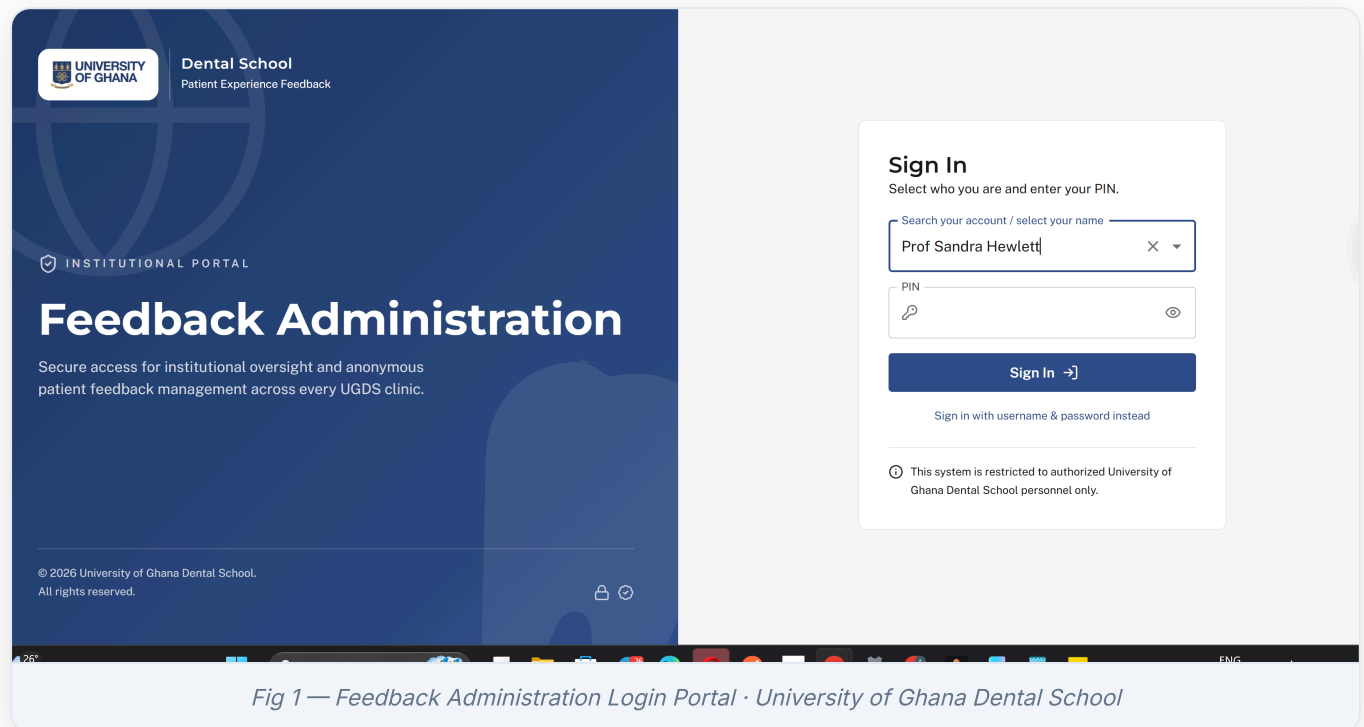
File Parser
UGDS_SMS — custom Windows folder-watcher service monitoring C:\eArchive_Bible\Old_Bible

Hosting
On-premises server, UGDS Korle Bu campus

3. Phase 1 Features — Live System

3.1 Secure Login Portal

Staff authenticate via name search + PIN, or username and password. Access is restricted to authorized UGDS personnel with role-based permissions.



- ▶ Staff name search + 4-digit PIN authentication
- ▶ Username & password fallback option
- ▶ Role-based access: Admin, Customer Care Officer, Unit Head
- ▶ UGDS institutional branding (University of Ghana crest & colours)
- ▶ Restricted to authorized UGDS personnel only

3.2 Overview Dashboard

The central command center. Gives leadership an instant hospital-wide view of patient satisfaction, feedback trends, and active complaints in real time.

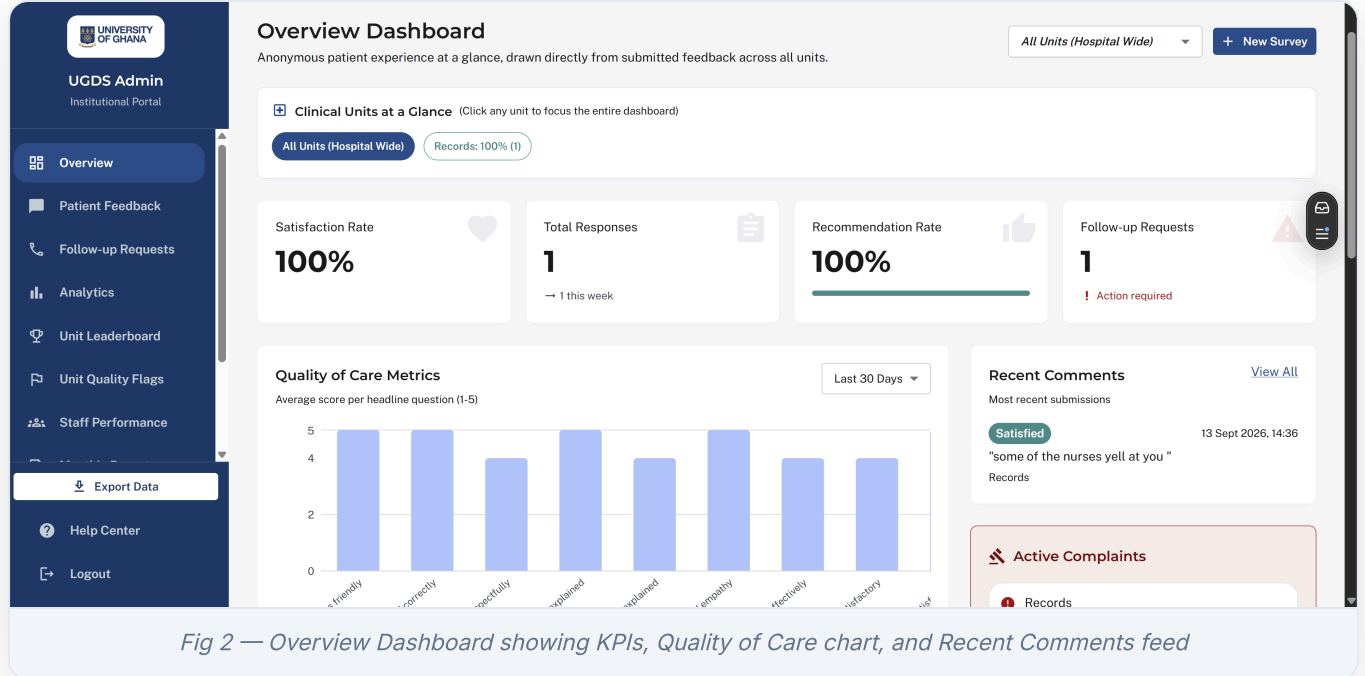


Fig 2 — Overview Dashboard showing KPIs, Quality of Care chart, and Recent Comments feed

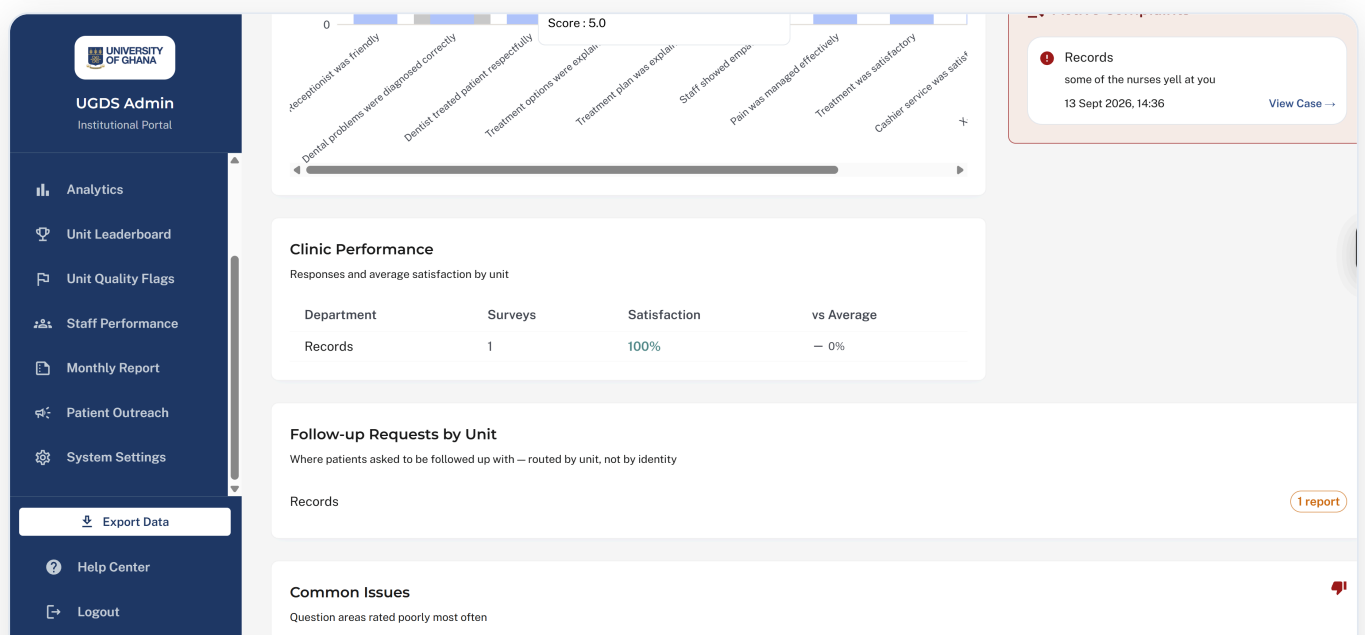
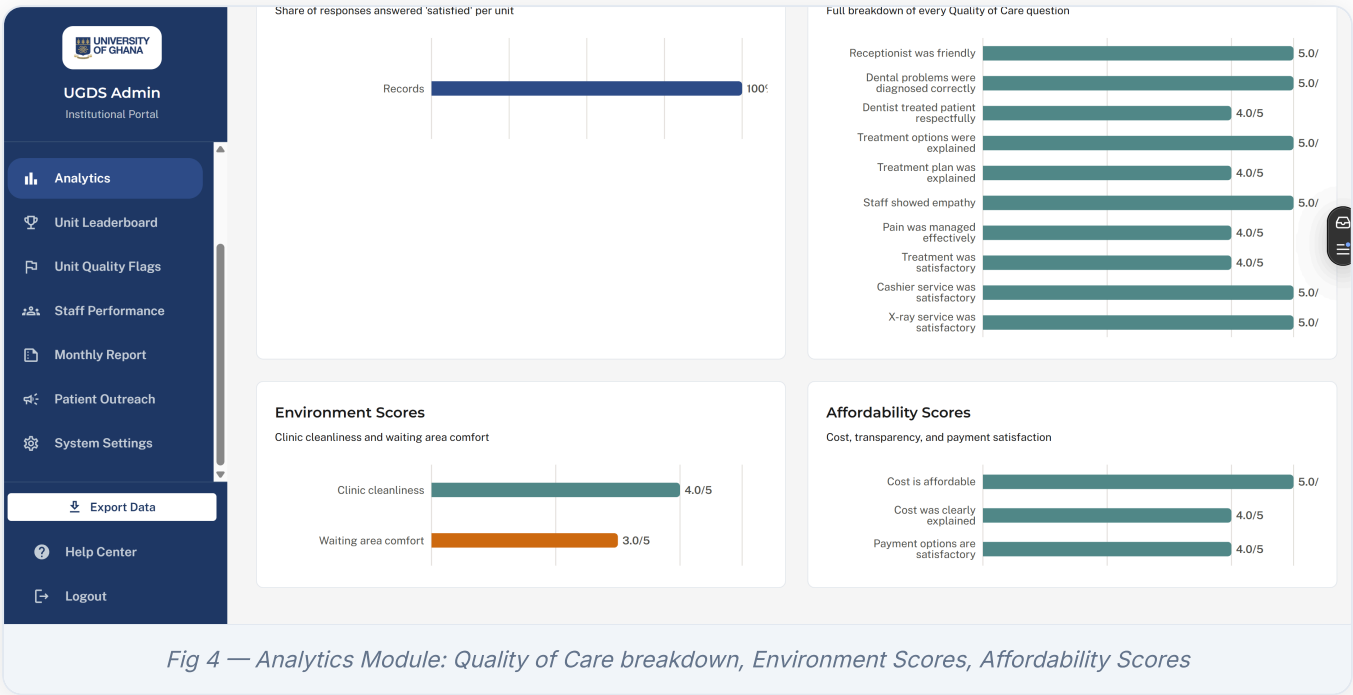


Fig 3 — Dashboard lower section: Clinic Performance table, Follow-up Requests, Common Issues, Navigation sidebar

- ▶ **Satisfaction Rate** — Live % of satisfied patients
- ▶ **Total Responses** — Running count of submitted surveys
- ▶ **Recommendation Rate** — Would patients recommend UGDS?
- ▶ **Follow-up Requests** — Flagged patients requiring a callback
- ▶ **Quality of Care Chart** — Bar chart scoring 9 clinical care dimensions
- ▶ **Recent Comments Feed** — Anonymous patient verbatim feedback (live)
- ▶ **Active Complaints Panel** — Red-flagged issues requiring urgent action
- ▶ **Clinic Performance Table** — Responses and satisfaction per unit
- ▶ **Unit Filter** — Click any unit to focus the entire dashboard

3.3 Analytics Module

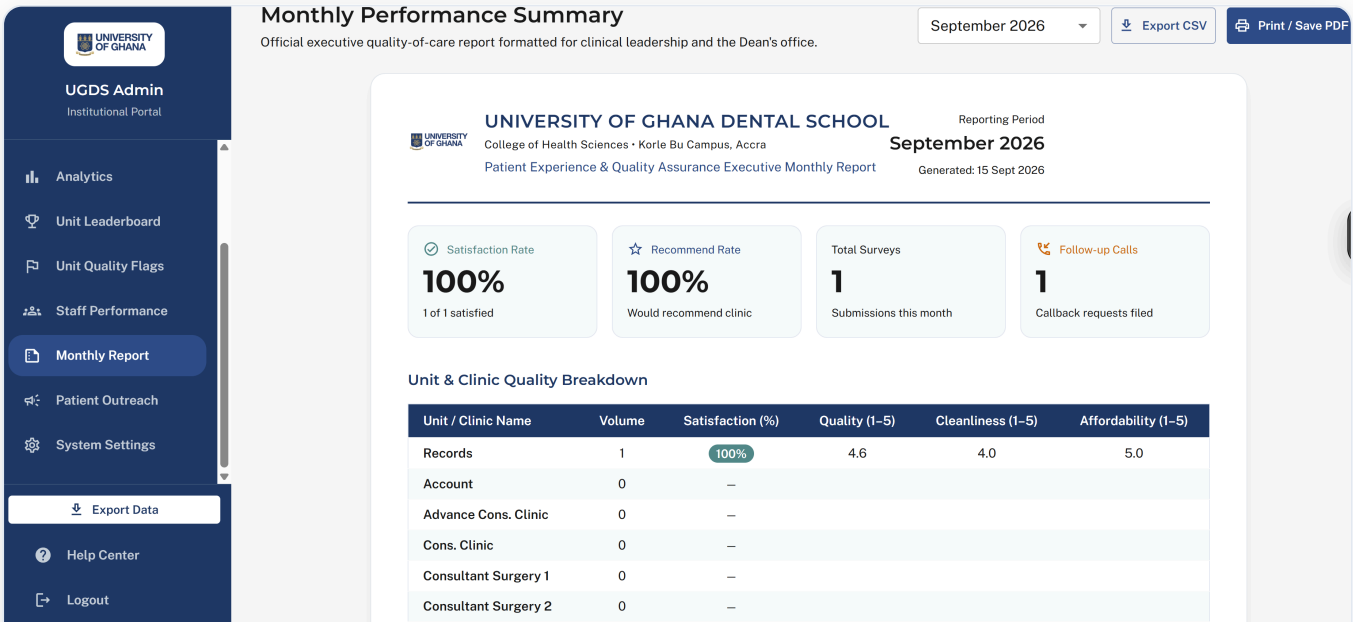
Deep-dive analysis of quality metrics across all clinical departments, broken down by care dimension, environment, and affordability.



- ▶ Satisfaction per Unit — horizontal bar chart with % scores per clinic
- ▶ Per-question scoring across 11 care dimensions (receptionist, diagnosis, empathy, pain management, etc.)
- ▶ Environment Scores — cleanliness & waiting area comfort
- ▶ Affordability Scores — cost transparency and payment satisfaction

3.4 Monthly Executive Report

Executive-grade reporting formatted for the Dean's office and clinical leadership — auto-generated with full institutional branding.



- ▶ Institutional letterhead with UGDS crest and College of Health Sciences branding
- ▶ Summary KPIs: Satisfaction Rate, Recommendation Rate, Total Surveys, Follow-up Calls
- ▶ Unit & Clinic Quality Breakdown table (Volume, Satisfaction %, Quality 1–5, Cleanliness 1–5, Affordability 1–5)
- ▶ All UGDS clinics listed (Records, Account, Advance Cons. Clinic, Cons. Clinic, Consultant Surgery 1 & 2, etc.)
- ▶ Export CSV + Print/Save PDF — month selector for historical comparison

3.5 Birthday Outreach Hub

Enables customer care officers to personally call patients on or around their birthday — the human touch alongside the automated birthday SMS.

- ▶ **Today's Celebrants** — Active list of patients celebrating today
- ▶ **Tomorrow's Preview** — One-day advance list for call preparation
- ▶ **This Week / This Month** — Extended outreach planning windows
- ▶ **"Call Personally" Button** — Officers initiate direct personal calls
- ▶ **Checkmark Button** — Mark completed calls for call logging and audit
- ▶ Automated birthday SMS sent separately via Arkesel — this hub manages the personal touch

3.6 Patients Directory

- ▶ Connected to **125,042 patient records** parsed from the eArchive system
- ▶ Search by name, patient ID, or phone number
- ▶ Visit status tracking (visited / not yet visited this month)
- ▶ Patient profile cards with last visit date
- ▶ Foundation for Phase 3's full Patient Management System

3.7 Unit Quality Flags

- ▶ Flags auto-raised when a unit's satisfaction score falls below threshold
- ▶ Visual red-flag indicators on the main dashboard
- ▶ Email notifications dispatched to unit heads and administrators
- ▶ Proactive quality management — no waiting for monthly reviews

3.8 Staff Performance Module

- ▶ Per-staff satisfaction scores derived from patient feedback
- ▶ Follow-up request attribution by staff member
- ▶ Complaint linkage to assigned staff
- ▶ Foundation for performance review and recognition programs

3.9 Unit Leaderboard

- ▶ All UGDS clinical units ranked by patient satisfaction score
- ▶ Weekly and monthly leaderboard snapshots
- ▶ Visible to all staff — drives healthy competition and service excellence

3.10 Patient Outreach SMS Engine

- ▶ Folder Watcher Service (UGDS_SMS) monitors C:\eArchive_Bible\Old_Bible for new eDump_*.txt files
- ▶ New patient record detected → SMS survey dispatched via Arkesel API v2 automatically
- ▶ Anonymous multi-question survey with 1–5 scale ratings
- ▶ Responses parsed and stored in PostgreSQL in real time
- ▶ Fully automated — zero manual effort required from staff

4. Phase 1 Delivery Summary

Module	Description	Status
Secure Login Portal	PIN + username/password, role-based access	✔ Complete
Overview Dashboard	Real-time KPIs, charts, complaints feed	✔ Complete
Patient Feedback Collection	Automated SMS surveys via Arkesel	✔ Complete
Follow-up Requests Module	Callback queue routed by unit	✔ Complete
Analytics Module	Multi-dimension quality scoring charts	✔ Complete
Unit Leaderboard	Ranked satisfaction scores by clinic	✔ Complete
Unit Quality Flags + Email Alerts	Auto-flag underperforming units	✔ Complete
Staff Performance Tracking	Per-staff satisfaction attribution	✔ Complete
Monthly Executive Report	PDF/CSV export, Dean's office ready	✔ Complete
Birthday Outreach Hub	Call-personally + checkmark logging	✔ Complete
Patients Directory	125,042 records, searchable index	✔ Complete
Export Data	CSV and PDF export across all modules	✔ Complete

5. Development Roadmap

PHASE 2 — PLANNED ENHANCEMENTS

Feature	Description
Email Alert System	Automated email to unit heads when quality flags are raised
Birthday Campaign Scheduling	Pre-scheduled campaigns with customizable message templates
Advanced Analytics	Month-over-month trends, predictive quality flags
Staff Login Portal	Individual staff dashboards with personal performance metrics

Feature	Description
In-App Notification Center	Real-time alerts for complaints, callbacks, and flags
Mobile Responsive Design	Full optimization for tablets and mobile phones

PHASE 3 — FULL HOSPITAL MANAGEMENT SYSTEM

Feature	Description
Full Patient Management System	Patient profiles, visit history, treatment records
Appointment Scheduling	Online and in-person booking management
Clinical Records Integration	Linking feedback to clinical outcomes
Lab & X-ray Management	Tracking diagnostic requests and results
Billing & Payments Module	Integrated fee management and receipts
Call Center Integration	Full CRM for the customer care team

6. Technical Glossary

Term	Definition
eArchive	UGDS's legacy patient record system stored as .txt dump files
eDump file	Auto-generated patient record file format (eDump_*.txt)
Arkesel	Ghana-based SMS gateway for survey and birthday messages
Neon PostgreSQL	Cloud-hosted relational database for all system data
FastAPI	Python web framework powering the system backend API
Vite + React	Modern JavaScript frontend framework for the admin portal
UGDS_SMS	Custom Windows folder-watcher service (locally built)
PIN Auth	4-digit PIN login method for quick, secure staff access

7. System Ownership

Role	Responsibility
Customer Care Unit	Day-to-day operation, birthday outreach calls, follow-up management
IT / Digital Systems	Backend maintenance, server uptime, SMS credit management
Unit Heads	Reviewing quality flags and responding to underperformance alerts
Dean's Office	Receiving and acting on monthly executive performance reports